

NIFTY 100 Analytics Platform

Analyst Guide

Purpose

This guide explains how an analyst can operate the Streamlit dashboard, use the company screener, review company and peer information, access reports, call the FastAPI endpoints, and troubleshoot common issues.

Current project components

The project contains an ETL/data-processing layer, financial analytics, a FastAPI REST API, a Streamlit dashboard, generated reports and tearsheets, and an automated pytest test suite.

Current verified project facts

- 92 master companies are represented in the processed company master.

1. Getting Started

Prerequisites

Use Python 3.12 in the current project environment. The project uses FastAPI/Uvicorn for the REST API and Streamlit for the dashboard.

Important packages used by the current project include pandas, FastAPI, Uvicorn, Streamlit, Plotly, pytest, pytest-html, SQLAlchemy and scikit-learn.

Project directory

```
C:\Users\saidh\OneDrive\Desktop\NIFTY100_ETL
```

Starting the API

From the project root:

```
uvicorn src.api.main:app --host 127.0.0.1 --port 8000
```

API documentation is available at /docs while the server is running.

Starting the dashboard

In another terminal:

```
streamlit run src\dashboard\app.py --server.port 8501
```

The dashboard is normally available at <http://localhost:8501>.

2. Dashboard Navigation

The current dashboard contains eight screens:

Screen	File	Purpose
Home	01_home.py	Main landing/overview screen
Company Profile	02_profile.py	Company financial profile and ratio history
Screener	03_screener.py	Filter and rank companies
Peers	04_peers.py	Peer-group analysis
Trends	05_trends.py	Financial trend analysis
Sectors	06_sectors.py	Sector-level analysis
Capital	07_capital.py	Capital allocation analysis
Reports	08_reports.py	Generated reports and documents

Analyst workflow

Start on Home, select a company when profile-level analysis is needed, use Screener for systematic filtering, then use Peers and Sectors to provide comparative context. Use Trends and Capital for deeper financial interpretation and Reports for generated outputs.

3. Company Profile

The Company Profile screen is implemented in `src/dashboard/pages/02_profile.py`.

Current workflow

1. The dashboard loads the company list and financial ratios.
2. Select a company from the Company selector.
3. The latest available financial snapshot is displayed.
4. A Net Profit Margin trend chart is shown.
5. Historical financial ratios are displayed in a table.
6. Company financial data can be downloaded as CSV.

Displayed snapshot fields

- Net Profit Margin
- Book Value / Share
- Dividend Payout
- Total Debt

Profile analysis

Use the profile screen to inspect whether profitability, balance-sheet metrics and historical ratios are stable over time. The trend chart is useful for identifying changes in net profit margin rather than relying only on the latest value.

4. Using the Screener

The Screener is designed to narrow the company universe using financial and valuation conditions.

Available API screening parameters

- min_roe — minimum return on equity
- max_de — maximum debt-to-equity
- min_fcf — minimum free cash flow
- sector — broad sector filter
- min_rev_cagr_5yr — minimum five-year revenue CAGR
- min_pat_cagr_5yr — minimum five-year PAT CAGR
- max_pe — maximum P/E

Example analyst screen

A quality-oriented screen can begin with Min ROE = 15. Additional leverage, cash-flow, growth or valuation conditions can then be applied to reduce the candidate set.

Interpretation

A screener is a filtering tool, not a substitute for company-level analysis. After identifying candidates, open the Company Profile and review the relevant historical ratios and trends.

5. Peers, Trends, Sectors and Capital

Peers

Use the Peers screen to compare companies within peer groups. The project also exposes peer-group and peer-comparison API routes.

Trends

The Trends screen provides historical financial context. Use it when the latest-year snapshot does not adequately explain the direction of a KPI.

Sectors

The current processed data contains 10 broad sectors: Communication Services, Consumer Discretionary, Consumer Staples, Energy, Financials, Healthcare, Industrials, Information Technology, Materials and Real Estate.

Note: the original project requirement specified 11 sectors, but the current source dataset contains 10. No artificial sector was added.

Capital

Use the Capital screen to review capital allocation and related financial characteristics produced by the analytics pipeline.

6. Reports and PDF Tearsheet Workflow

The project generates company tearsheets as PDF reports. The report-generation code is under `src/reports/`.

Current tearsheet location

`reports/tearsheets/`

The current project directory contains 89 tearsheet PDFs. Earlier project acceptance criteria specified 92 tearsheets, so this discrepancy should be resolved before final sign-off.

Using a tearsheet

Open the PDF for the required company and review the financial profile, trend information and generated analytical content. Tearsheets are intended as compact company-level reports for analyst review.

Selected report outputs

Generated outputs also include screener results, valuation summaries, peer comparisons, portfolio statistics, cluster labels, cluster profiles, cash-flow analytics, outlier reports and pros/cons analysis.

7. FastAPI REST API

Base URL

```
http://127.0.0.1:8000/api/v1
```

Main route groups

- companies
- screener
- sectors
- peers
- valuation
- portfolio
- documents
- health

Health check

```
curl http://127.0.0.1:8000/api/v1/health
```

Company list

```
curl http://127.0.0.1:8000/api/v1/companies
```

Single company

```
curl http://127.0.0.1:8000/api/v1/companies/TCS
```

Screener

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15"
```

Invalid company IDs return HTTP 404. Invalid screener parameter values are designed to return HTTP 400.

8. API Data and Operational Examples

Company financial history

P&L::

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/pl
```

Balance sheet:

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/bs
```

Cash flow:

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/cashflow
```

Ratios:

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/ratios
```

Documents:

```
curl http://127.0.0.1:8000/api/v1/companies/TCS/documents
```

Portfolio statistics:

```
curl http://127.0.0.1:8000/api/v1/portfolio/stats
```

Interactive API documentation

With FastAPI running, open /docs in a browser to inspect the generated OpenAPI documentation and try endpoints interactively.

9. Testing and Performance

Run the full test suite from the project root:

```
python -m pytest tests\ -v
```

Generate an HTML report:

```
python -m pytest tests\ --html=reports\pytest_report.html --self-contained-html
```

Performance testing

Day 43 includes a concurrent screener load test using 10 Python threads and a Company Profile performance check for five tickers.

The screener load test targets completion of all 10 concurrent requests within 10 seconds. The Company Profile data-loading test targets less than 3 seconds per ticker.

SQLite optimization

Indexes were added for frequently filtered company/year combinations in `financial_ratios`, `profitandloss`, `balancesheet`, `cashflow`, `market_cap` and `financial_metrics`, plus company indexes for `sectors` and `peer_groups`.

10. Troubleshooting

FastAPI command not found

Use the Python module form if required:

```
python -m uvicorn src.api.main:app --port 8000
```

Streamlit command not found

Install Streamlit and use the module form:

```
python -m pip install streamlit  
python -m streamlit run src\dashboard\app.py
```

Missing Plotly

Install Plotly:

```
python -m pip install plotly
```

SQLite table missing

Inspect tables with sqlite3 and confirm that the required analytics tables exist. The project currently includes a financial_metrics table loaded from data/processed/financial_metrics.csv.

API test returns 422 instead of 400

Screener numeric query parameters are converted and validated inside the endpoint so invalid values can return HTTP 400.

11. Data Outputs and Analyst Checklist

Important generated outputs

The output directory contains analytics such as cluster_labels.csv, cluster_profiles.csv, cashflow_intelligence.xlsx, portfolio_stats.csv, outlier_report.csv, pros_cons_generated.csv, peer_comparison.xlsx, valuation_summary.xlsx and screener_output.xlsx.

Final analyst checklist

1. Start FastAPI and confirm /api/v1/health returns status=ok.
2. Start Streamlit and confirm the dashboard opens.
3. Use Screener with appropriate filters.
4. Open Company Profile for shortlisted companies.
5. Review peer and sector context.
6. Review generated reports/tearsheets.
7. Run the test suite before publishing results.
8. Check generated output files and validation reports.

Known finalization items

The current project snapshot reports 89 PDFs in reports/tearsheets/, while the earlier acceptance criterion requires 92. Resolve this before declaring the final deliverable set complete.

End of Analyst Guide